

Q1

- (a) Derive Poisson's and Laplace's equations from the point form of Gauss's Law.
- (b) Calculate the spherical capacitance of two conducting concentric spheres of radii a and b , $b > a$ using Laplacian operator for one dimensional change in potential, $V(r)$.
Boundary conditions are given ahead.

$r = a$	$V = 0$
$r = b$	$V = V_0$

- (c) Determine the potential energy stored in the spherical capacitors by two methods:

i)
$$W_E = \frac{1}{2} \int_v \epsilon E^2 dv$$

ii)
$$W_E = \frac{1}{2} QV_0$$

National University of Computer and Emerging Sciences, Lahore Campus



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Instruction/Notes:

Question#1. [10]

Given a $90\mu\text{C}$ point charge located at the origin, find the total electric flux passing through a sphere $r = 26\text{cm}$. First plot the surface, guess the answer and then find the required flux.

Question#2. [15]

A sphere of radius ' a ' has a uniform volume charge density of $\rho_o \text{ C/m}^3$. Using Gauss's law, find the electric flux intensity inside and outside the sphere.

Question#3. [10+5=15]

- (a) A total charge of $Q = \frac{40}{3} \text{ nC}$ is uniformly distributed in the form of a circular disk of radius 2m. Find the potential due to this charge at a point on the axis, 2m from the disk.
- (b) Again find the potential if same Q is located at the origin and compare it with (a).

Question#4. [2+2+2+2+2=10]

The potential in free space is given by, $V = 2xy^2z^3 + 3 \ln(x^2 + 2y^2 + 3z^2) \text{ V}$. Evaluate each of the following quantities at $P(3, 2, -1)$:

- (a) V
- (b) $\vec{E}$
- (c) $\vec{D}$
- (d) $\hat{a}_N$
- (e) ρ_v

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Q2

- (a) Define electric dipole. A dipole moment of $p = 6a_z$ [nC m] is located at the origin in free-space. Find (i) V at $P(r = 5, \theta = 30^\circ, \phi = 0^\circ)$ and (ii) E at P .
- (b) Four point charges $Q_1 = +q$, $Q_2 = -q$, $Q_3 = -q$, and $Q_4 = +q$ are located at $(0,0)$, $(a,0)$, $(0,a)$, and (a,a) respectively. Draw diagram and completely work out for the potential energy stored in this quadrupole.
- (c) A homogeneous dielectric ($\epsilon_r = 2.5$) fills region 1 ($x \leq 0$) while region 2 ($x \geq 0$) is free space. Find D_2 and θ_2 , if $D_1 = 12a_x - 10a_y + 4a_z$ [nC/m²]

CLO # 03 and 04